INSY7213

Assignment 1

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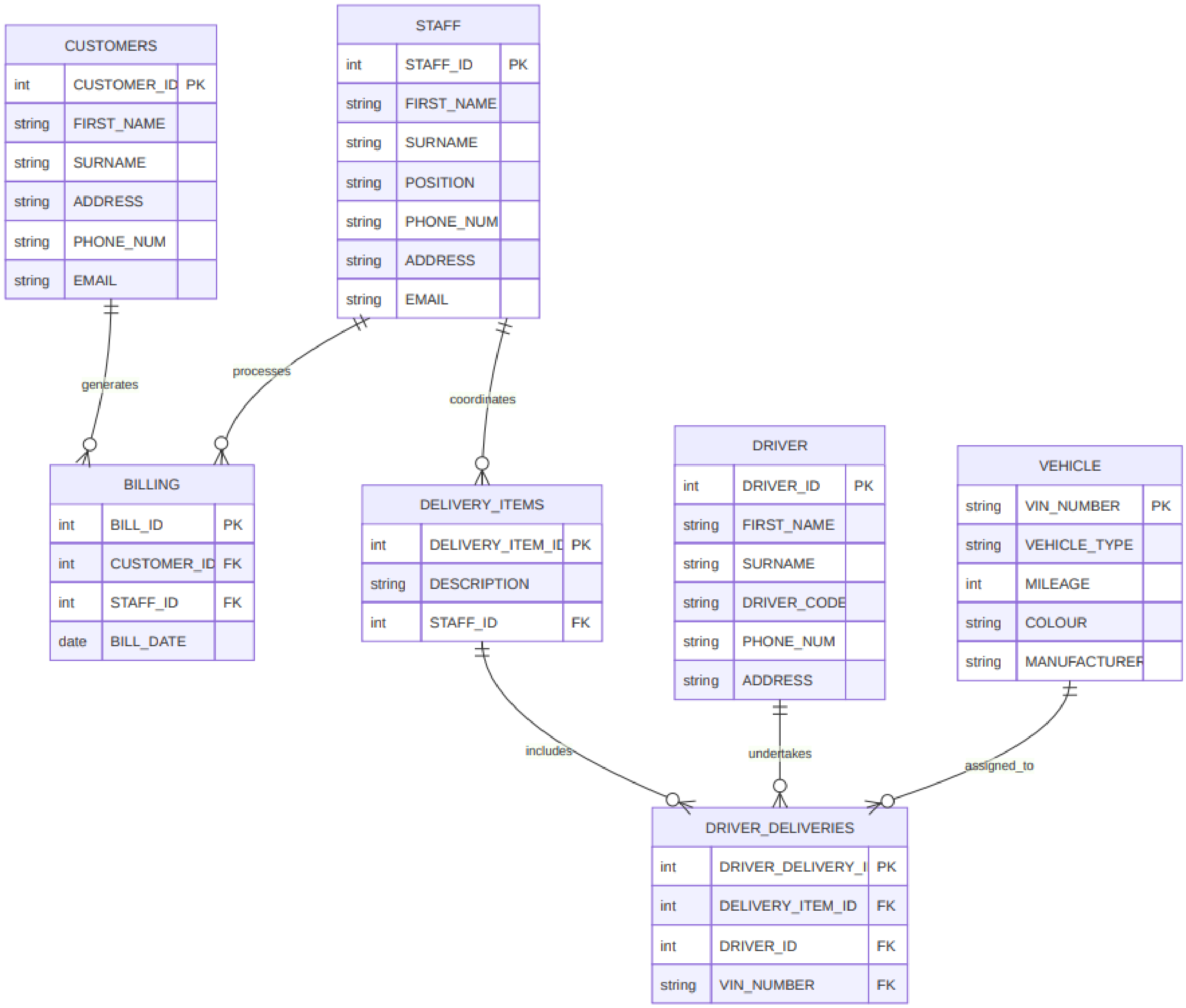
QUESTION 1


CUSTOMERS
int CUSTOMER_ID PK
string FIRST_NAME
string SURNAME
string ADDRESS
string PHONE_NUM
string EMAIL
STAFF
int STAFF_ID PK
string FIRST_NAME
string SURNAME
string POSITION
string PHONE_NUM
string ADDRESS
string EMAIL
generates
processes
coordinates
BILLING
int BILL_ID PK
int CUSTOMER_ID FK
int STAFF_ID FK
date BILL_DATE
DELIVERY_ITEMS
int DELIVERY_ITEM_ID PK
string DESCRIPTION
int STAFF_ID FK
DRIVER
int DRIVER_ID PK
string FIRST_NAME
string SURNAME
string DRIVER_CODE
string PHONE_NUM
string ADDRESS
VEHICLE
string VIN_NUMBER PK
string VEHICLE_TYPE
int MILEAGE
string COLOUR
string MANUFACTURER
includes
undertakes
assigned_to
DRIVER_DELIVERIES
int DRIVER_DELIVERY_I PK
int DELIVERY_ITEM_ID FK
int DRIVER_ID FK
string VIN_NUMBER FK

QUESTION 2

Creating the tables and inputting the data

```
CREATE TABLE CUSTOMERS (
    CUSTOMER_ID NUMBER(10) PRIMARY KEY,
    FIRST_NAME  VARCHAR2(50)  NOT NULL,
    SURNAME     VARCHAR2(50)  NOT NULL,
    ADDRESS     VARCHAR2(150),
    PHONE_NUM   VARCHAR2(20),
    EMAIL       VARCHAR2(100)
);

CREATE TABLE STAFF (
    STAFF_ID    NUMBER(10) PRIMARY KEY,
    FIRST_NAME  VARCHAR2(50)  NOT NULL,
    SURNAME     VARCHAR2(50)  NOT NULL,
    POSITION    VARCHAR2(50),
    PHONE_NUM   VARCHAR2(20),
    ADDRESS     VARCHAR2(150),
    EMAIL       VARCHAR2(100)
);

CREATE TABLE DRIVER (
    DRIVER_ID   NUMBER(10) PRIMARY KEY,
    FIRST_NAME  VARCHAR2(50)  NOT NULL,
    SURNAME     VARCHAR2(50)  NOT NULL,
    DRIVER_CODE VARCHAR2(10)  NOT NULL,
    PHONE_NUM   VARCHAR2(20),
    ADDRESS     VARCHAR2(150)
);

CREATE TABLE VEHICLE (
    VIN_NUMBER   VARCHAR2(30) PRIMARY KEY,
    VEHICLE_TYPE VARCHAR2(50),
    MILEAGE      NUMBER(10),
    COLOUR       VARCHAR2(30),
```

```
  MANUFACTURER VARCHAR2(50)
);

CREATE TABLE BILLING (
  BILL_ID    NUMBER(10) PRIMARY KEY,
  CUSTOMER_ID NUMBER(10) NOT NULL,
  STAFF_ID   NUMBER(10) NOT NULL,
  BILL_DATE  DATE NOT NULL,
  CONSTRAINT FK_BILLING_CUSTOMER FOREIGN KEY (CUSTOMER_ID) REFERENCES
CUSTOMERS(CUSTOMER_ID),
  CONSTRAINT FK_BILLING_STAFF FOREIGN KEY (STAFF_ID) REFERENCES STAFF(STAFF_ID)
);

CREATE TABLE DELIVERY_ITEMS (
  DELIVERY_ITEM_ID NUMBER(10) PRIMARY KEY,
  DESCRIPTION    VARCHAR2(150) NOT NULL,
  STAFF_ID       NUMBER(10) NOT NULL,
  CONSTRAINT FK_DELIV_STAFF FOREIGN KEY (STAFF_ID) REFERENCES STAFF(STAFF_ID)
);

CREATE TABLE DRIVER_DELIVERIES (
  DRIVER_DELIVERY_ID NUMBER(10) PRIMARY KEY,
  VIN_NUMBER       VARCHAR2(30) NOT NULL,
  DRIVER_ID        NUMBER(10) NOT NULL,
  DELIVERY_ITEM_ID  NUMBER(10) NOT NULL,
  CONSTRAINT FK_DD_VIN FOREIGN KEY (VIN_NUMBER) REFERENCES VEHICLE(VIN_NUMBER),
  CONSTRAINT FK_DD_DRIVER FOREIGN KEY (DRIVER_ID) REFERENCES DRIVER(DRIVER_ID),
  CONSTRAINT FK_DD_DELIV_ITEM FOREIGN KEY (DELIVERY_ITEM_ID) REFERENCES
DELIVERY_ITEMS(DELIVERY_ITEM_ID)
);

-- INSERTING DATA (DML)

-- CUSTOMERS
INSERT INTO CUSTOMERS VALUES (11011, 'Bob', 'Smith', '18 Water rd', '0877277521', 'bobs@isat.com');
INSERT INTO CUSTOMERS VALUES (11012, 'Sam', 'Hendricks', '22 Water rd', '0863257857',
'shen@mcom.co.za');
```

INSERT INTO CUSTOMERS VALUES (11013, 'Larry', 'Clark', '101 Summer lane', '0834567891', 'larc@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11014, 'Jeff', 'Jones', '55 Mountain way', '0612547895', 'jj@isat.co.za');
INSERT INTO CUSTOMERS VALUES (11015, 'Andre', 'Kerk', '5 Main rd', '0827238521', 'akerk@mcal.co.za');
INSERT INTO CUSTOMERS VALUES (11016, 'Wayne', 'Smith', '13 Water rd', '0877277522', 'ws@isat.com');
INSERT INTO CUSTOMERS VALUES (11017, 'John', 'Hendricks', '29 Water rd', '0863257851', 'jhen@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11018, 'Sally', 'Clark', '111 Summer lane', '0834567892', 'sallyc@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11019, 'Bridget', 'Bitterhour', '125 Mountain way', '0612547896', 'bb@isat.co.za');
INSERT INTO CUSTOMERS VALUES (11111, 'Nicole', 'Kerk', '175 Main rd', '0827238529', 'nk@mcal.co.za');
INSERT INTO CUSTOMERS VALUES (11112, 'Catherine', 'Smith', '19 Water rd', '0877277523', 'cath@isat.com');
INSERT INTO CUSTOMERS VALUES (11113, 'Mel', 'Hendricks', '5 Water rd', '0863257852', 'melh@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11114, 'Lucy', 'Du Plessis', '221 Summer lane', '0834567892', 'ldup@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11116, 'Josh', 'Maverick', '155 Mountain way', '0612547897', 'joshm@isat.co.za');
INSERT INTO CUSTOMERS VALUES (11117, 'Stuart', 'Jones', '35 Main rd', '0827238521', 'sjones@mcal.co.za');
-- STAFF
INSERT INTO STAFF VALUES (51011, 'Sally', 'Du Toit', 'Logistics', '0825698547', '18 Main rd', 'sdut@isat.com');
INSERT INTO STAFF VALUES (51012, 'Mark', 'Wright', 'CRM', '0836984178', '12 Cape Way', 'mwright@isat.com');
INSERT INTO STAFF VALUES (51013, 'Harry', 'Sheen', 'Logistics', '0725648965', '15 Water Street', 'hsheen@isat.com');
INSERT INTO STAFF VALUES (51014, 'Jabu', 'Xolani', 'Logistics', '0823116598', '18 White Lane', 'jxo@isat.com');
INSERT INTO STAFF VALUES (51015, 'Roberto', 'Henry', 'Packaging', '0783521451', '55 Cape Street', 'rhenry@isat.com');
INSERT INTO STAFF VALUES (51016, 'Pat', 'Durant', 'Logistics', '0825698542', '1 Main rd', 'pd@isat.com');
INSERT INTO STAFF VALUES (51017, 'Steve', 'Maritz', 'CRM', '0836984173', '2 Cape Way', 'sm@isat.com');
INSERT INTO STAFF VALUES (51018, 'Maxwell', 'Dube', 'Logistics', '0725648964', '5 Water Street', 'max@isat.com');
INSERT INTO STAFF VALUES (51019, 'Shane', 'Mane', 'Logistics', '0823116595', '8 White Lane', 'smane@isat.com');

```sql
INSERT INTO STAFF VALUES (51111, 'Bob', 'Truth', 'Packaging', '0783521456', '35 Cape Street', 'btruth@isat.com');

-- DRIVER
INSERT INTO DRIVER VALUES (81011, 'Buthelezi', 'Marshall', 'C1', '0725698547', '18 Leopard creek');
INSERT INTO DRIVER VALUES (81012, 'Tina', 'Mtati', 'C', '0636984178', '12 Cape rd');
INSERT INTO DRIVER VALUES (81013, 'Jono', 'Mvuyisi', 'EC1', '0725648965', '15 Circle lane');
INSERT INTO DRIVER VALUES (81014, 'Richard', 'Smith', 'C1', '0623116598', '18 Beach rd');
INSERT INTO DRIVER VALUES (81015, 'Brett', 'Smith', 'EB', '0883521457', '55 Summer lane');

-- VEHICLE
INSERT INTO VEHICLE VALUES ('1ZA55858541', 'Cutaway van chassis', 115352, 'RED', 'MAN');
INSERT INTO VEHICLE VALUES ('1ZA51858542', 'Flatbed truck', 315856, 'BLUE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA35858543', 'Medium Standard Truck', 789587, 'SILVER', 'MAN');
INSERT INTO VEHICLE VALUES ('1ZA15851545', 'Flatbed truck', 555050, 'WHITE', 'TATA');
INSERT INTO VEHICLE VALUES ('1ZA35868540', 'Cutaway van chassis', 79058, 'WHITE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA65858541', 'Cutaway van chassis', 215352, 'RED', 'MAN');
INSERT INTO VEHICLE VALUES ('1ZA85868540', 'Flatbed truck', 215856, 'BLUE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA55858543', 'Medium Standard Truck', 889587, 'SILVER', 'MERC');
INSERT INTO VEHICLE VALUES ('1ZA65851545', 'Flatbed truck', 155050, 'WHITE', 'MAN');
INSERT INTO VEHICLE VALUES ('1ZA65868540', 'Cutaway van chassis', 19058, 'WHITE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA75858541', 'Cutaway van chassis', 315352, 'RED', 'MAN');
INSERT INTO VEHICLE VALUES ('1ZA71858542', 'Flatbed truck', 115856, 'BLUE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA75858543', 'Medium Standard Truck', 989587, 'SILVER', 'MERC');
INSERT INTO VEHICLE VALUES ('1ZA17851545', 'Flatbed truck', 755050, 'WHITE', 'TATA');
INSERT INTO VEHICLE VALUES ('1ZA75868540', 'Cutaway van chassis', 29058, 'WHITE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA85858541', 'Cutaway van chassis', 515352, 'RED', 'MERC');
INSERT INTO VEHICLE VALUES ('1ZA81858542', 'Flatbed truck', 715856, 'BLUE', 'ISUZU');
INSERT INTO VEHICLE VALUES ('1ZA85858543', 'Medium Standard Truck', 789587, 'SILVER', 'MAN');
INSERT INTO VEHICLE VALUES ('1ZA85851545', 'Flatbed truck', 555050, 'WHITE', 'TATA');
INSERT INTO VEHICLE VALUES ('1ZA85868540', 'Cutaway van chassis', 39058, 'WHITE', 'MERC');

-- BILLING
INSERT INTO BILLING VALUES (800, 11011, 51011, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (801, 11012, 51013, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (802, 11014, 51015, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
```

```sql
INSERT INTO BILLING VALUES (803, 11015, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (804, 11013, 51014, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (805, 11111, 51011, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (806, 11012, 51013, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (807, 11014, 51015, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (808, 11015, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (809, 11113, 51018, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (810, 11011, 51011, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (811, 11012, 51013, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (812, 11014, 51016, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (813, 11117, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (814, 11013, 51014, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (815, 11012, 51111, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (816, 11012, 51019, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (817, 11014, 51015, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (818, 11112, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (819, 11013, 51014, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (820, 11116, 51019, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));

-- DELIVERY_ITEMS
INSERT INTO DELIVERY_ITEMS VALUES (71011, 'House relocation', 51011);
INSERT INTO DELIVERY_ITEMS VALUES (71012, 'Delivery of consignments', 51017);
INSERT INTO DELIVERY_ITEMS VALUES (71013, 'Delivery of specialized consignments', 51015);
INSERT INTO DELIVERY_ITEMS VALUES (71014, 'Office relocation', 51012);
INSERT INTO DELIVERY_ITEMS VALUES (71015, 'Delivery of specialized consignments', 51014);

-- DRIVER_DELIVERIES
INSERT INTO DRIVER_DELIVERIES VALUES (91011, '1ZA55858541', 81011, 71011);
INSERT INTO DRIVER_DELIVERIES VALUES (91012, '1ZA35858543', 81012, 71013);
INSERT INTO DRIVER_DELIVERIES VALUES (91013, '1ZA17851545', 81011, 71015);
INSERT INTO DRIVER_DELIVERIES VALUES (91014, '1ZA35868540', 81013, 71015);
INSERT INTO DRIVER_DELIVERIES VALUES (91015, '1ZA15851545', 81014, 71012);

COMMIT;
```

RESULTS

Table CUSTOMERS created.
Table STAFF created.
Table DRIVER created.
Table VEHICLE created.
Table BILLING created.
Table DELIVERY_ITEMS created.
Table DRIVER_DELIVERIES created.
1 row inserted.
1 row inserted.
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1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
Commit complete.
Table DRIVER_DELIVERIES dropped.
Table DELIVERY_ITEMS dropped.
Table BILLING dropped.
Table VEHICLE dropped.
Table DRIVER dropped.
Table STAFF dropped.
Table CUSTOMERS dropped.
Table CUSTOMERS created.
Table STAFF created.
Table DRIVER created.
Table VEHICLE created.
Table BILLING created.
Table DELIVERY_ITEMS created.
Table DRIVER_DELIVERIES created.
1 row inserted.
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1 row inserted.
1 row inserted.
Commit complete.

QUESTION 3

```
-- Switch to the pluggable database (Standard Oracle XE 18c/21c architecture)
ALTER SESSION SET CONTAINER = XEPDB1;

-- 1. Create User John
CREATE USER John IDENTIFIED BY Johnch2026
   DEFAULT TABLESPACE users
   TEMPORARY TABLESPACE temp
   QUOTA UNLIMITED ON users;

-- Grant Connection and SELECT ANY TABLE Privilege
GRANT CREATE SESSION TO John;
GRANT SELECT ANY TABLE TO John;

-- 2. Create User Hannah
CREATE USER Hannah IDENTIFIED BY Hannahch2026
   DEFAULT TABLESPACE users
   TEMPORARY TABLESPACE temp
   QUOTA UNLIMITED ON users;

-- Grant Connection and INSERT ANY TABLE Privilege
GRANT CREATE SESSION TO Hannah;
GRANT INSERT ANY TABLE TO Hannah;
```

Results

Session altered.
User JOHN created.
Grant succeeded.
Grant succeeded.
User HANNAH created.
Grant succeeded.
Grant succeeded.

QUESTION 4

```sql
-- 1. Create and populate DRIVER
CREATE TABLE DRIVER (
   DRIVER_ID   NUMBER(10) PRIMARY KEY,
   FIRST_NAME  VARCHAR2(50)  NOT NULL,
   SURNAME     VARCHAR2(50)  NOT NULL,
   DRIVER_CODE VARCHAR2(10)  NOT NULL,
   PHONE_NUM   VARCHAR2(20),
   ADDRESS     VARCHAR2(150)
);

INSERT INTO DRIVER VALUES (81011, 'Buthelezi', 'Marshall', 'C1', '0725698547', '18 Leopard creek');
INSERT INTO DRIVER VALUES (81012, 'Tina', 'Mtati', 'C', '0636984178', '12 Cape rd');
INSERT INTO DRIVER VALUES (81013, 'Jono', 'Mvuyisi', 'EC1', '0725648965', '15 Circle lane');
INSERT INTO DRIVER VALUES (81014, 'Richard', 'Smith', 'C1', '0623116598', '18 Beach rd');
INSERT INTO DRIVER VALUES (81015, 'Brett', 'Smith', 'EB', '0883521457', '55 Summer lane');

-- 2. Create and populate DRIVER_DELIVERIES
CREATE TABLE DRIVER_DELIVERIES (
   DRIVER_DELIVERY_ID NUMBER(10) PRIMARY KEY,
   VIN_NUMBER       VARCHAR2(30) NOT NULL,
   DRIVER_ID        NUMBER(10) NOT NULL,
   DELIVERY_ITEM_ID   NUMBER(10) NOT NULL
);

INSERT INTO DRIVER_DELIVERIES VALUES (91011, '1ZA55858541', 81011, 71011);
INSERT INTO DRIVER_DELIVERIES VALUES (91012, '1ZA35858543', 81012, 71013);
INSERT INTO DRIVER_DELIVERIES VALUES (91013, '1ZA17851545', 81011, 71015);
INSERT INTO DRIVER_DELIVERIES VALUES (91014, '1ZA35868540', 81013, 71015);
INSERT INTO DRIVER_DELIVERIES VALUES (91015, '1ZA15851545', 81014, 71012);

COMMIT;
```

```
SET SERVEROUTPUT ON;

DECLARE
   CURSOR c_low_mileage IS
      SELECT
         d.FIRST_NAME,
         d.SURNAME,
         d.DRIVER_CODE,
         dd.VIN_NUMBER,
         v.MILEAGE
      FROM DRIVER_DELIVERIES dd
      JOIN DRIVER d ON dd.DRIVER_ID = d.DRIVER_ID
      JOIN VEHICLE v ON dd.VIN_NUMBER = v.VIN_NUMBER
      WHERE v.MILEAGE < 80000;
BEGIN
   FOR rec IN c_low_mileage LOOP
      DBMS_OUTPUT.PUT_LINE('--------------------------');
      DBMS_OUTPUT.PUT_LINE('DRIVER: ' || rec.FIRST_NAME || ', ' || rec.SURNAME);
      DBMS_OUTPUT.PUT_LINE('CODE: ' || rec.DRIVER_CODE);
      DBMS_OUTPUT.PUT_LINE('VIN NUMBER: ' || rec.VIN_NUMBER);
      DBMS_OUTPUT.PUT_LINE('MILEAGE: ' || rec.MILEAGE);
   END LOOP;
   DBMS_OUTPUT.PUT_LINE('--------------------------');
END;
/
```

Results

```
--------------------------
DRIVER: Jono, Mvuyisi
CODE: EC1
VIN NUMBER: 1ZA35868540
MILEAGE: 79058
--------------------------
```

```
PL/SQL procedure successfully completed.
```

QUESTION 5

Q.5.1

```
-- CREATE AND POPULATE STAFF & DELIVERY_ITEMS

CREATE TABLE STAFF (
   STAFF_ID    NUMBER(10) PRIMARY KEY,
   FIRST_NAME  VARCHAR2(50)  NOT NULL,
   SURNAME     VARCHAR2(50)  NOT NULL,
   POSITION    VARCHAR2(50),
   PHONE_NUM   VARCHAR2(20),
   ADDRESS     VARCHAR2(150),
   EMAIL       VARCHAR2(100)
);

INSERT INTO STAFF VALUES (51011, 'Sally', 'Du Toit', 'Logistics', '0825698547', '18 Main rd', 'sdut@isat.com');
INSERT INTO STAFF VALUES (51012, 'Mark', 'Wright', 'CRM', '0836984178', '12 Cape Way', 'mwright@isat.com');
INSERT INTO STAFF VALUES (51013, 'Harry', 'Sheen', 'Logistics', '0725648965', '15 Water Street', 'hsheen@isat.com');
INSERT INTO STAFF VALUES (51014, 'Jabu', 'Xolani', 'Logistics', '0823116598', '18 White Lane', 'jxo@isat.com');
INSERT INTO STAFF VALUES (51015, 'Roberto', 'Henry', 'Packaging', '0783521451', '55 Cape Street', 'rhenry@isat.com');
INSERT INTO STAFF VALUES (51016, 'Pat', 'Durant', 'Logistics', '0825698542', '1 Main rd', 'pd@isat.com');
INSERT INTO STAFF VALUES (51017, 'Steve', 'Maritz', 'CRM', '0836984173', '2 Cape Way', 'sm@isat.com');
INSERT INTO STAFF VALUES (51018, 'Maxwell', 'Dube', 'Logistics', '0725648964', '5 Water Street', 'max@isat.com');
INSERT INTO STAFF VALUES (51019, 'Shane', 'Mane', 'Logistics', '0823116595', '8 White Lane', 'smane@isat.com');
INSERT INTO STAFF VALUES (51111, 'Bob', 'Truth', 'Packaging', '0783521456', '35 Cape Street', 'btruth@isat.com');

CREATE TABLE DELIVERY_ITEMS (
   DELIVERY_ITEM_ID NUMBER(10) PRIMARY KEY,
   DESCRIPTION      VARCHAR2(150) NOT NULL,
   STAFF_ID         NUMBER(10) NOT NULL
```

```sql
);

INSERT INTO DELIVERY_ITEMS VALUES (71011, 'House relocation', 51011);
INSERT INTO DELIVERY_ITEMS VALUES (71012, 'Delivery of consignments', 51017);
INSERT INTO DELIVERY_ITEMS VALUES (71013, 'Delivery of specialized consignments', 51015);
INSERT INTO DELIVERY_ITEMS VALUES (71014, 'Office relocation', 51012);
INSERT INTO DELIVERY_ITEMS VALUES (71015, 'Delivery of specialized consignments', 51014);

COMMIT;
```

```sql
SET SERVEROUTPUT ON;

DECLARE
  v_staff_id   STAFF.STAFF_ID%TYPE;
  v_first_name STAFF.FIRST_NAME%TYPE;
  v_surname    STAFF.SURNAME%TYPE;
  v_deliveries NUMBER;
BEGIN
  SELECT
    s.STAFF_ID,
    s.FIRST_NAME,
    s.SURNAME,
    COUNT(dd.DRIVER_DELIVERY_ID)
  INTO
    v_staff_id,
    v_first_name,
    v_surname,
    v_deliveries
  FROM STAFF s
  JOIN DELIVERY_ITEMS di ON s.STAFF_ID = di.STAFF_ID
  JOIN DRIVER_DELIVERIES dd ON di.DELIVERY_ITEM_ID = dd.DELIVERY_ITEM_ID
  GROUP BY s.STAFF_ID, s.FIRST_NAME, s.SURNAME
  ORDER BY COUNT(dd.DRIVER_DELIVERY_ID) DESC
  FETCH FIRST 1 ROW ONLY;

```

```
   DBMS_OUTPUT.PUT_LINE('--------------------------');
   DBMS_OUTPUT.PUT_LINE('STAFF ID: ' || v_staff_id);
   DBMS_OUTPUT.PUT_LINE('FIRST NAME: ' || v_first_name);
   DBMS_OUTPUT.PUT_LINE('SURNAME: ' || v_surname);
   DBMS_OUTPUT.PUT_LINE('DELIVERIES PROCESSED: ' || v_deliveries);
   DBMS_OUTPUT.PUT_LINE('--------------------------');
END;
/
```

Results

```
Table STAFF created.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
Table DELIVERY_ITEMS created.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
Commit complete.

STAFF ID: 51014
FIRST NAME: Jabu
SURNAME: Xolani
DELIVERIES PROCESSED: 2

PL/SQL procedure successfully completed.
```

Q.5.2

1. **Declarative :** This is the start, where everything gets initialized, it has the variables and everything needed in the sql. In Q.5.1, it declares v_staff_id, v_first_name, v_surname, and v_deliveries using %TYPE anchoring.

2. **BEGIN :**This is the logic, for the program. The data input and its minitulation, the loops and its other conditions for the queries. This is what gives the results or what we are looking to find. In Q.5.1, this joins the tables, aggregates counts, stores the result, and outputs it with DBMS_OUTPUT.

3. **Exception Handling :** This section handles errors that happened during the program, where there was syntax or logic errors. Its needed to make sure that the SQL will function and if there was a problem it will be fixed. NO_DATA_FOUND, TOO_MANY_ROWS)

Q.5.3

Q.5.3.1

A View is a virtual table defined by a stored SQL query. It does not store physical copies of data itself; instead, it runs its underlying query whenever called.

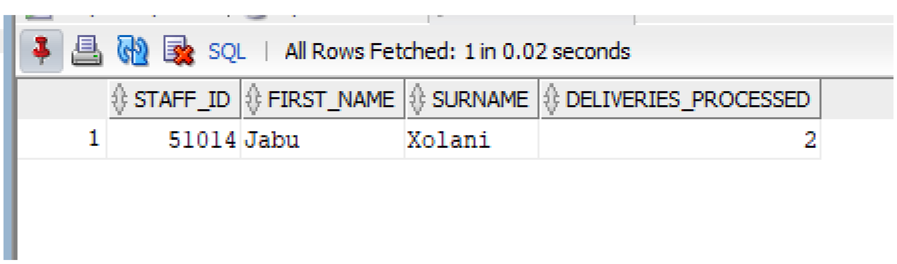
A view can be useful, because it lets non technical people have a grasp of the data and be able to pull data and have it shown in a way that they can understand.

Q5.3.2

```
-- Step 1: Create the view to calculate delivery totals per staff member
CREATE OR REPLACE VIEW VW_STAFF_DELIVERY_SUMMARY AS
SELECT
   s.STAFF_ID,
   s.FIRST_NAME,
   s.SURNAME,
   -- Count every completed delivery handled by this staff member
   COUNT(dd.DRIVER_DELIVERY_ID) AS DELIVERIES_PROCESSED
FROM STAFF s
-- Connect the staff member to the delivery catalog items they arranged
JOIN DELIVERY_ITEMS di ON s.STAFF_ID = di.STAFF_ID
-- Connect those items to actual driver road deliveries
JOIN DRIVER_DELIVERIES dd ON di.DELIVERY_ITEM_ID = dd.DELIVERY_ITEM_ID
-- Group rows together so each staff member gets one combined total
GROUP BY s.STAFF_ID, s.FIRST_NAME, s.SURNAME;

-- Step 2: Query the view to find the highest-performing staff member
SELECT
   STAFF_ID,
   FIRST_NAME,
   SURNAME,
   DELIVERIES_PROCESSED
FROM VW_STAFF_DELIVERY_SUMMARY
ORDER BY DELIVERIES_PROCESSED DESC
FETCH FIRST 1 ROW ONLY;
```

Results



	STAFF_ID	FIRST_NAME	SURNAME	DELIVERIES_PROCESSED
1	51014	Jabu	Xolani	2

QUESTION 6

Q.6.1

1

Implicit Cursor

Implicit Cursors do not need to declare the curser. With The implicit Curser we can track information about the curser. They give the results from attributes like INSERT, UPDATE, DELETE, ETC. it makes use of different attributes as well like %FOUND, %NOTFOUND and %ROWCOUNT

Explicit Cursor

Explicit Cursors are defined and declared by the programers so that there is more control on the context area. They are usually defined with the SELECT Statement. It is a pointer that moves though the data and the programers are the ones which make it start, move and stop. Declared and managed manually by the developer for precise row-by-row processing over multi-row queries. Use attributes like c_name%NOTFOUND or c_name%ROWCOUNT to manage loop conditions and verify data presence.

2

```
SET SERVEROUTPUT ON;

-- 1. Explicit Cursor Attribute Demo (%NOTFOUND)
DECLARE
  -- Explicitly define cursor to retrieve all Mercedes (MERC) vehicles
  CURSOR c_merc_vehicles IS
    SELECT VIN_NUMBER, VEHICLE_TYPE, MILEAGE
    FROM VEHICLE
    WHERE MANUFACTURER = 'MERC';

  v_vin     VEHICLE.VIN_NUMBER%TYPE;
  v_type    VEHICLE.VEHICLE_TYPE%TYPE;
  v_mileage VEHICLE.MILEAGE%TYPE;
BEGIN
  OPEN c_merc_vehicles;
  DBMS_OUTPUT.PUT_LINE('--- MERCEDES FLEET LIST ---');
  LOOP
    FETCH c_merc_vehicles INTO v_vin, v_type, v_mileage;
    -- Explicit cursor attribute: exit when no more records exist
    EXIT WHEN c_merc_vehicles%NOTFOUND;

```

```
    DBMS_OUTPUT.PUT_LINE('VIN: ' || v_vin || ' | Type: ' || v_type || ' | Mileage: ' || v_mileage);
  END LOOP;
  CLOSE c_merc_vehicles;
END;
/

-- 2. Implicit Cursor Attribute Demo (SQL%ROWCOUNT)
BEGIN
  -- Update vehicle mileage for testing
  UPDATE VEHICLE
  SET MILEAGE = MILEAGE + 500
  WHERE MANUFACTURER = 'TATA';

  -- Implicit cursor attribute: confirms how many rows were modified
  IF SQL%FOUND THEN
    DBMS_OUTPUT.PUT_LINE('Update successful. Total vehicles updated (SQL%ROWCOUNT): ' ||
SQL%ROWCOUNT);
  ELSE
    DBMS_OUTPUT.PUT_LINE('No records matched update criteria.');
  END IF;

  -- Rollback test update to maintain baseline data
  ROLLBACK;
END;
/
```

Q6.2

Sequences generate unique, auto-incrementing integers independent of table data. In Cheetah Deliveries, manually typing IDs causes primary key collisions and requires race-condition-prone queries like MAX(BILL_ID) + 1. A sequence ensures concurrent transactions get unique numbers with minimal lock contention.

```
-- 1. Create and populate CUSTOMERS (needed for Foreign Key reference)
CREATE TABLE CUSTOMERS (
  CUSTOMER_ID NUMBER(10) PRIMARY KEY,
  FIRST_NAME  VARCHAR2(50)  NOT NULL,
  SURNAME     VARCHAR2(50)  NOT NULL,
```

```sql
   ADDRESS    VARCHAR2(150),
   PHONE_NUM  VARCHAR2(20),
   EMAIL      VARCHAR2(100)
);

INSERT INTO CUSTOMERS VALUES (11011, 'Bob', 'Smith', '18 Water rd', '0877277521', 'bobs@isat.com');
INSERT INTO CUSTOMERS VALUES (11012, 'Sam', 'Hendricks', '22 Water rd', '0863257857', 'shen@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11013, 'Larry', 'Clark', '101 Summer lane', '0834567891', 'larc@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11014, 'Jeff', 'Jones', '55 Mountain way', '0612547895', 'jj@isat.co.za');
INSERT INTO CUSTOMERS VALUES (11015, 'Andre', 'Kerk', '5 Main rd', '0827238521', 'akerk@mcal.co.za');
INSERT INTO CUSTOMERS VALUES (11016, 'Wayne', 'Smith', '13 Water rd', '0877277522', 'ws@isat.com');
INSERT INTO CUSTOMERS VALUES (11017, 'John', 'Hendricks', '29 Water rd', '0863257851', 'jhen@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11018, 'Sally', 'Clark', '111 Summer lane', '0834567892', 'sallyc@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11019, 'Bridget', 'Bitterhour', '125 Mountain way', '0612547896', 'bb@isat.co.za');
INSERT INTO CUSTOMERS VALUES (11111, 'Nicole', 'Kerk', '175 Main rd', '0827238529', 'nk@mcal.co.za');
INSERT INTO CUSTOMERS VALUES (11112, 'Catherine', 'Smith', '19 Water rd', '0877277523', 'cath@isat.com');
INSERT INTO CUSTOMERS VALUES (11113, 'Mel', 'Hendricks', '5 Water rd', '0863257852', 'melh@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11114, 'Lucy', 'Du Plessis', '221 Summer lane', '0834567892', 'ldup@mcom.co.za');
INSERT INTO CUSTOMERS VALUES (11116, 'Josh', 'Maverick', '155 Mountain way', '0612547897', 'joshm@isat.co.za');
INSERT INTO CUSTOMERS VALUES (11117, 'Stuart', 'Jones', '35 Main rd', '0827238521', 'sjones@mcal.co.za');

-- 2. Create and populate BILLING
CREATE TABLE BILLING (
   BILL_ID     NUMBER(10) PRIMARY KEY,
   CUSTOMER_ID NUMBER(10) NOT NULL,
   STAFF_ID    NUMBER(10) NOT NULL,
   BILL_DATE   DATE NOT NULL,
   CONSTRAINT FK_BILLING_CUSTOMER FOREIGN KEY (CUSTOMER_ID) REFERENCES CUSTOMERS(CUSTOMER_ID),
```

```sql
  CONSTRAINT FK_BILLING_STAFF FOREIGN KEY (STAFF_ID) REFERENCES STAFF(STAFF_ID)
);

INSERT INTO BILLING VALUES (800, 11011, 51011, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (801, 11012, 51013, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (802, 11014, 51015, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (803, 11015, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (804, 11013, 51014, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (805, 11111, 51011, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (806, 11012, 51013, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (807, 11014, 51015, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (808, 11015, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (809, 11113, 51018, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (810, 11011, 51011, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (811, 11012, 51013, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (812, 11014, 51016, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (813, 11117, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (814, 11013, 51014, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (815, 11012, 51111, TO_DATE('06-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (816, 11012, 51019, TO_DATE('07-SEP-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (817, 11014, 51015, TO_DATE('10-NOV-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (818, 11112, 51012, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (819, 11013, 51014, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));
INSERT INTO BILLING VALUES (820, 11116, 51019, TO_DATE('09-DEC-2022', 'DD-MON-YYYY'));

COMMIT;
```

```sql
-- 1. Insert new record using the sequence
INSERT INTO BILLING (BILL_ID, CUSTOMER_ID, STAFF_ID, BILL_DATE)
VALUES (SEQ_BILL_ID.NEXTVAL, 11011, 51011, SYSDATE);

COMMIT;

-- 2. Query the inserted record
SELECT
```

BILL_ID,
CUSTOMER_ID,
STAFF_ID,
TO_CHAR(BILL_DATE, 'DD-Mon-YYYY HH24:MI:SS') AS BILL_TIMESTAMP
FROM BILLING
WHERE BILL_ID = (SELECT MAX(BILL_ID) FROM BILLING);

Results

Table CUSTOMERS created.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
Table BILLING created.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.

1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
1 row inserted.
Commit complete.
1 row inserted.
Commit complete.
BILL_ID CUSTOMER_ID   STAFF_ID BILL_TIMESTAMP

```
   BILL_ID CUSTOMER_ID   STAFF_ID BILL_TIMESTAMP
---------- ----------- ---------- -----------------------------
       821       11011      51011 07-Sep-2026 13:58:48
```

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